

Variational Contraction Conditions for Iterative Algorithms in Multi-Population Discrete-Time Regularized Mean-Field Games

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Abstract—In this work, we study the contraction conditions of iterative algorithms for stationary and finite-horizon discrete-time regularized mean-field games (MFGs) with multiple populations, where each population only interacts with the state distributions of the other populations. Due to the high dimensionality caused by the interaction of different populations, contraction rates for these algorithms cannot, in general, be expressed in terms of radicals. By studying the dynamics of these iterative algorithms and assuming that the system components of each population’s MFG are Lipschitz continuous, we present explicit (eventual) contraction conditions for each algorithm in any normed space, relying only on these Lipschitz parameters. As a consequence of these contraction conditions, we provide convergence rates of finite-horizon mean-field equilibria to infinite-horizon stationary (and nonstationary) mean-field equilibria (MFEs), under restrictions on a variational characterization of the dynamics of these iterative algorithms. In the single-population case, the restrictions we impose on this variational characterization to obtain these convergence results are less restrictive than previous results in the literature.

I. INTRODUCTION

In this work, we study the convergence of some basic iterative algorithms in the setting of discrete-time regularized mean-field games with multiple populations. We assume that interactions between populations are *weak*, in the sense that populations interact with each other only through their state distributions, which is a standard setting in the literature on multi-population MFGs [1], [2], [3]. Our main goal is to identify dynamical systems associated with these iterative algorithms and to study the contraction properties of these dynamical systems, which imply global asymptotic stability of the mean-field equilibrium (MFE). Beyond deterministic analysis, these global asymptotic stability properties also enable the use of basic learning algorithms when the components of the MFG are unknown, such as stochastic approximation [4].

In model-based learning settings [5], [6], most methods are restricted to finite-horizon problems. In the single-population case, [7, Theorem 4] shows that learning a finite-horizon MFG with a sufficiently large horizon length also suffices for the infinite-horizon setting under additional assumptions on the system [7, Corollaries 5 and 6]. Another application of [7, Theorem 4] is a uniqueness condition for infinite-horizon non-stationary MFEs obtained via finite-horizon

MFEs, which is not directly available when studying the infinite-horizon system itself [7, Proposition 3]. For these reasons, it is desirable to obtain contraction conditions in the multi-population setting, especially in the finite-horizon case, since each of these results relies on establishing a horizon-length-independent contraction condition for finite-horizon MFGs.

The main difficulty in establishing contraction conditions for multi-population MFGs is the cross-interaction among populations. This cross-interaction, especially in the finite-horizon case, leads to a high-dimensional dynamical system, which complicates the analysis. In our case, to study contraction in these high-dimensional systems under Lipschitz continuity assumptions, we identify a matrix whose spectral radius (which, in this case, is an eigenvalue) characterizes the contraction property of the MFG. Since eigenvalues are roots of characteristic polynomials, closed-form expressions in radicals are often unavailable by the Abel–Ruffini theorem. Moreover, the matrices arising in the finite-horizon case are dense and lack exploitable structure; as a result, the runtime of brute-force eigenvalue computation scales with the matrix dimension. In particular, with N populations and horizon length T , this requires computing the spectral radius of an $N(T - 1) \times N(T - 1)$ matrix. Finally, due to cross-interactions, the proofs used to obtain contraction results in the single-population case do not readily extend to the multi-population setting. Therefore, easily verifiable contraction conditions are especially desirable for multi-population MFGs.

A. Literature Review

This work primarily extends the results of [7] to the multi-population setting. The main distinction between our work and [7] is that the proofs in [7] do not readily extend to the multi-population case. The contraction condition in [7, Theorem 2] relies on arguments that are purely algebraic (with the exception of [7, Lemma 9], which also admits a purely algebraic proof in the setting of that work). In this work, we transfer the eigenvector problem associated with the matrix characterizing contraction to the complex plane using generating functions, and we argue using complex-analytic methods common in the analytic combinatorics literature.

In the discrete-time multi-population MFGs there are very few works available. The work [3] studies basic RL algorithms in the setting of LQR average-cost MFGs in discrete-time. Although contraction is used for the learning algorithms, the work [3, Theorem 3] does not provide an easy to check con-

Research of the authors was supported in part by the AFOSR Grant FA9550-24-1-0152.

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dition under which the proposed contraction condition holds. In [2], existence of MFE is studied for multi-population discrete-time MFGs in the general case. The work [8] applies model-free learning on stationary multi-population discrete-time MFGs that occur in networks. The work [9] also models a network problem as a multi-population discrete-time average-cost MFG and uses supervised-learning to learn the MFE, for which a blanket contraction assumption is used. In [1], a multi-population MFG model is proposed to study epidemics in continuous time, where contraction arguments are used for learning the MFE. We note that all these works also consider weak interaction between populations.

B. Contributions

- 1) In Section III, we present a contraction condition for a quasi-static iteration algorithm for multi-population stationary MFGs under regularization (Theorem 2), where one solves an MDP as an intermediate step. This result, in particular, extends [10, Theorem 1] to the multi-population setting.
- 2) In Section IV, we present *variational contraction conditions* for multi-population finite-horizon MFGs under regularization (Theorem 4), which is the main result of this work. The variational contraction condition we provide reduces the problem of computing the spectral radius of the associated matrix to a one-dimensional optimization problem. In particular cases—for instance, in the single-population setting—the contraction result we establish is equivalent to [7, Theorem 2]. Thus, Theorem 4 extends [7, Theorem 2] to the multi-population setting.
- 3) In Section V, we present rates of convergence and finite-time error bounds between the finite-horizon MFE and the infinite-horizon MFE (Theorem 5). The variational contraction condition we establish allows us to obtain convergence-rate results even in cases not covered by [7, Theorem 4] in the single-population setting. In particular, our variational contraction condition enables us to establish a convergence rate between the finite-horizon MFE and the infinite-horizon MFE whenever the stationary MFE is contractive, which is not always possible under [7, Theorem 4].

II. PRELIMINARIES

In the mean-field setting, a *population* is a continuum of homogeneous players. In our multi-population mean-field setting, we assume that there is a large number of players, each belonging to a specific population. Each population has its own reward function and transition kernel.

Formally, a discrete-time *multi-population mean-field game (MFG)* is described as follows:

- 1) The game is played in discrete time, $t \in \{1, 2, \dots\}$.
- 2) The game is played by a continuum of players divided into N populations. For simplicity, we assume that all populations share the same (finite) state space X and (finite) action space A .

- 3) A family of state measures $\tau_t = (\tau_{t,1}, \tau_{t,2}, \dots, \tau_{t,N}) \in \prod_{i=1}^N \mathcal{P}(X)$ denotes the global state of the game at time t , where $\tau_{t,i}$ is the state distribution of population i . We assume that at each stage t , every player observes their own state, their population state measure, and the state measures of the other populations. Thus, the interaction between populations is *weak*.
- 4) The one-stage cost for an individual in population i is given by a function $c_i : X \times A \times \prod_{j=1}^N \mathcal{P}(X) \rightarrow [0, \infty)$. The quantity $c_i(x, a, \tau)$ is the cost incurred at a stage when the player's state is x , the action is a , and the joint population state distribution is τ .
- 5) State transitions for an individual in population i are defined by the stochastic kernel $p_i : X \times A \times \prod_{j=1}^N \mathcal{P}(X) \rightarrow \mathcal{P}(X)$.

With the conventions above, we denote a multi-population MFG by the tuple $(X, A, (c_i)_{i \in [N]}, (p_i)_{i \in [N]}, N)$. Thus, a multi-population MFG can be viewed as a collection of N distinct discrete-time MFGs coupled through the populations' state distributions. The main distinction between a multi-population MFG and a single-population MFG is the cross-interaction among different populations.

For a given $t \in \mathbb{N}$ and $i \in [N]$, a *policy* is a stochastic kernel $\pi_{t,i} : X \rightarrow \mathcal{P}(A)$, which specifies the action distribution of population i at time t conditional on the current state. We denote by Π the space of policies $(\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$. The fact that policies depend only on an agent's own state (and not directly on the states of other populations) is an artifact of weak interaction.

In general, a discrete-time MFG is an extension of discrete-time Markov decision processes (MDPs) [11]. Indeed, an MFG can be reduced to an MDP by fixing a family of state measures $(\tau_t)_{t=0}^\infty$. Let $(\beta_i)_{i \in [N]} \subset (0, 1)$. In the infinite-horizon case, defining an objective function for each population requires additional structure.

For each $i \in [N]$, $\beta_i < 1$ is the discount factor used by population i . Under these discount factors, and for a given family of multi-population state measures $\tau = (\tau_t)_{t=0}^\infty$, the infinite-horizon objective function of the population $i \in [N]$ is defined as

$$J_i(\pi, \tau) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \beta_i^t c_i(x_t, a_t, \tau_{t,1}, \tau_{t,2}, \dots, \tau_{t,N}) \right],$$

where $\pi = (\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$.

We say that a family of policies $\pi := (\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$ is *optimal* for a family of state measures $\tau := (\tau_{t,i})_{t \in \mathbb{N}, i \in [N]}$ if

$$\begin{aligned} & \tau_{t+1,i}(\cdot) \\ &= \int_X \int_A p_i(\cdot \mid x, a, \tau_{t,1}, \dots, \tau_{t,N}) \pi_{t,i}(da \mid x) \tau_{t,i}(dx) \end{aligned} \quad (1)$$

$$(2)$$

for all $i \in [N]$. Conversely, we say that a family of state measures τ is *optimal* for a family of policies π if

$$J_i(\pi, \tau) = \inf_{\tilde{\pi} \in \Pi} J_i(\tilde{\pi}, \tau)$$

for all $i \in [N]$.

A solution to the MFG consists of a family of policies $(\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$ and a family of state measures $(\tau_{t,i})_{t \in \mathbb{N}, i \in [N]}$ such that $(\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$ is optimal for $(\tau_{t,i})_{t \in \mathbb{N}, i \in [N]}$ and $(\tau_{t,i})_{t \in \mathbb{N}, i \in [N]}$ is optimal for $(\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$. In this case, we call $((\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}, (\tau_{t,i})_{t \in \mathbb{N}, i \in [N]})$ a *mean-field equilibrium* (MFE).

In this work, we are interested in computing MFEs through iterative algorithms. For this purpose, we mainly work with *deterministic policies*. A policy $\pi = (\pi_{t,i})_{t \in \mathbb{N}, i \in [N]} \in \Pi$ is said to be *deterministic* if, for all $t \in \mathbb{N}$ and $i \in [N]$, there exists a function $f_{t,i} : X \rightarrow A$ such that $\pi_{t,i}(\cdot | x) = \delta_{f_{t,i}(x)}(\cdot)$, where δ denotes the Dirac measure. The space of all deterministic policies is denoted by Π_d .

To compute an MFE via iterative algorithms, we need to ensure that, throughout the iterations, there exists a unique Lipschitz policy. To achieve this, we first lift the action space of the MFG to $\mathcal{P}(A)$. This amounts to defining an MFG $(X, \mathcal{P}(A), (C_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$ that shares the “same” MFE as $(X, A, (c_i)_{i \in [N]}, (p_i)_{i \in [N]}, N)$. First, we embed $\mathcal{P}(A)$ into $\mathbb{R}^{|A|}$. For any $u \in \mathcal{P}(A)$, define

$$C_i(x, u, \tau_1, \dots, \tau_N) = \sum_{a \in A} c_i(x, a, \tau_1, \dots, \tau_N) u(a),$$

and

$$P_i(\cdot | x, u, \tau_1, \dots, \tau_N) = \sum_{a \in A} p_i(\cdot | x, a, \tau_1, \dots, \tau_N) u(a).$$

To ensure uniqueness and Lipschitz continuity of the policies that arise during the iterations, we regularize each C_i by adding a term $\rho_i \Omega_i$, where $\rho_i > 0$ and $\Omega_i : \mathcal{P}(A) \rightarrow \mathbb{R}$. We further assume that each Ω_i is 1-strongly convex with respect to the ℓ_1 -norm, i.e.,

$$\Omega_i(\theta u + (1-\theta)v) \leq \theta \Omega_i(u) + (1-\theta) \Omega_i(v) - \frac{1}{2} \theta(1-\theta) \|u - v\|_1^2$$

for all $u, v \in \mathcal{P}(A)$ and $\theta \in [0, 1]$. The resulting MFG $(X, \mathcal{P}(A), (C_i + \rho_i \Omega_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$ is called a *regularized MFG*, and the parameters ρ_i are called *regularization parameters*. The MFE of the regularized MFG generally differs from that of the original lifted MFG; however, the added regularization ensures the existence of a unique Lipschitz policy that optimizes the objective for any given state-measure trajectory. In what follows, we will suppose that we are working with a fixed regularized MFG $(X, \mathcal{P}(A), (C_i + \rho_i \Omega_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$.

The algorithms that we consider in this work will closely mimic the so-called value-iteration algorithm for the MDPs. For a given family of state measures $\tau \in \prod_{i=1}^N \mathcal{P}(X)$, we update the Q -functions as in the value-iteration algorithm. In order to satisfy the condition (2), we must also control the evolution of the state measures. First, we define a domain for the Q -functions for the population $i \in [N]$ as

$$\tilde{\mathcal{C}}_i := \left\{ Q^{\text{disc}} : X \times A \rightarrow \mathbb{R}_+ : \|Q\|_\infty \leq \frac{M}{1-\beta}, \right. \\ \left. |Q(x, a) - Q(\tilde{x}, \tilde{a})| \leq \frac{L_1}{1 - \frac{\beta K_1}{2}} (1_{x \neq \tilde{x}} + 2 \cdot 1_{a \neq \tilde{a}}) \right\}.$$

Then we define the set $\mathcal{C}_i$ as Q -functions over $X \times \mathcal{P}(A)$ such that $Q(x, u) := \sum_{a \in A} Q^{\text{disc}}(x, a) u(a) + \rho_i \Omega_i$. To handle the iterations of Q -functions and state-measures, we define the following operators:

Definition 1: Let $Q \in \mathcal{C}_i$ and $\tau \in \prod_{i=1}^N \mathcal{P}(X)$. For the updates of our Q -functions, we define the operator

$$H_{1,i}(Q, \tau)(x, u) \\ = C_i(x, u, \tau) + \rho_i \Omega_i(u) + \beta_i \sum_{y \in X} Q_{\min}(y) P_i(y | x, u, \tau),$$

and for the updates of our state measures we define

$$H_{2,i}(Q, \tau)(\cdot) = \sum_{x \in X} \int_{\mathcal{P}(A)} P_i(\cdot | x, u, \tau) \pi_Q(du | x) \tau_i(x),$$

where π_Q is an optimal policy induced by the Q -function Q .

The operator $H_{1,i}$ arises from the dynamic programming structure associated with the objective functions J_i , a well-known result in the theory of MDPs. The operator $H_{2,i}$ is specific to the MFG setting and reflects the cross-interaction required between policies and state measures. From the definition of $H_{1,i}$, note that C_i and P_i are affine over $\mathcal{P}(A)$ and Ω_i is strongly convex; therefore, for any given Q -function, $H_{1,i}$ admits a unique minimizer. In particular, the operator $H_{2,i}$ is well defined.

In this work, we consider three different types of MFE. In the infinite-horizon case, an MFE (π, τ) is said to be *stationary* if it is time independent, i.e., for all $i \in [N]$ and $t \in \mathbb{N}$,

$$\tau_{t,i} = \tau_{t+1,i} \quad \text{and} \quad \pi_{t,i} = \pi_{t+1,i}.$$

Due to this stationarity condition, we do not impose any restriction on the initial state measures when considering stationary MFEs. When initial state measures are not fixed, one typically seeks a stationary MFE.

Given an initial family of state measures $\tau_0 = (\tau_{0,i})_{i \in [N]} \in \prod_{i \in [N]} \mathcal{P}(X)$, we say that an MFE $(\pi, (\tilde{\tau}_{t,i})_{t \in \mathbb{N}, i \in [N]})$ is *non-stationary* (starting from τ_0) if $\tilde{\tau}_{0,i} = \tau_{0,i}$ for all $i \in [N]$. In the non-stationary case, the initial state measures are fixed *a priori*, typically reflecting the populations’ initial distributions. For $\tau_0 \in \prod_{i \in [N]} \mathcal{P}(X)$, we denote by MFE_{τ_0} the set of MFEs for the infinite-horizon MFG with states measure that start from τ_0 .

In the finite-horizon case, one often operates as in the infinite-horizon case, with the caveat that the objective function has the following form

$$J_i(\pi, \tau) = E_\pi \left(\sum_{t=0}^{T-1} \beta_i^t c_i(x_t, a_t, \tau_{t,1}, \tau_{t,2}, \dots, \tau_{t,N}) \right).$$

The termination of the iterations under the expectation above at time T forces a natural boundary condition to the iterations of the Q -functions. In this case, by MFE_{T, τ_0} , we will denote the set of finite-horizon MFEs starting from τ_0 .

To complete our setup, we impose the following Lipschitz continuity assumption on the system components c_i and p_i .

Assumption 1: Let $\|\cdot\|_1$ denote the ℓ_1 -norm on $\mathcal{P}(X) \subset \mathbb{R}^{|X|}$. For all $i \in [N]$, $x, \tilde{x} \in X$, $a, \tilde{a} \in \mathcal{P}(A)$, and

$$\tau = (\tau_j)_{j \in [N]}, \quad \tilde{\tau} = (\tilde{\tau}_j)_{j \in [N]} \in \prod_{j \in [N]} \mathcal{P}(X),$$

we have

$$\begin{aligned} & |c_i(x, a, \tau) - c_i(\tilde{x}, \tilde{a}, \tilde{\tau})| \\ & \leq L_i \left(\mathbf{1}_{\{x \neq \tilde{x}\}} + 2 \mathbf{1}_{\{a \neq \tilde{a}\}} + \sum_{j=1}^N \|\tau_j - \tilde{\tau}_j\|_1 \right), \end{aligned}$$

and

$$\begin{aligned} & \|p_i(\cdot | x, a, \tau) - p_i(\cdot | \tilde{x}, \tilde{a}, \tilde{\tau})\|_1 \\ & \leq K_i \left(\mathbf{1}_{\{x \neq \tilde{x}\}} + 2 \mathbf{1}_{\{a \neq \tilde{a}\}} + \sum_{j=1}^N \|\tau_j - \tilde{\tau}_j\|_1 \right). \end{aligned}$$

Lipschitz continuity assumptions of this type are common in the analysis of iterative algorithms.

We note that although we defined the operators $H_{1,i}$ and $H_{2,i}$ for the lifted system, the assumption above is stated for the original system. A similar condition also holds for the lifted system [10].

Lemma 1: For all $x, \tilde{x} \in X$, $u, \tilde{u} \in \mathcal{P}(A)$, and $\tau, \tilde{\tau} \in \prod_{i \in [N]} \mathcal{P}(X)$, the functions C_i and P_i satisfy the following Lipschitz bounds:

$$\begin{aligned} & |C_i(x, u, \tau) - C_i(\tilde{x}, \tilde{u}, \tilde{\tau})| \\ & \leq L_i \left(\mathbf{1}_{\{x \neq \tilde{x}\}} + \|u - \tilde{u}\|_1 + \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1 \right), \end{aligned}$$

and

$$\begin{aligned} & \|P_i(\cdot | x, u, \tau) - P_i(\cdot | \tilde{x}, \tilde{u}, \tilde{\tau})\|_1 \\ & \leq K_i \left(\mathbf{1}_{\{x \neq \tilde{x}\}} + \|u - \tilde{u}\|_1 + \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1 \right). \end{aligned}$$

Proof: The proof is the same as in [10]. ■

An important consequence of Lemma 1 is the following Lipschitz property for $H_{1,i}$ and $H_{2,i}$, which will be fundamental in obtaining a variational expression for our iterations in the stationary and finite-horizon settings. Let $\bar{K}_i = \frac{3}{2}K_i + \frac{1}{2} \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i)}$.

Lemma 2: For any $Q, \tilde{Q} \in \mathcal{C}_i$ and $\tau, \tilde{\tau} \in \prod_{i=1}^N \mathcal{P}(X)$, we have

$$\begin{aligned} & \|H_{1,i}(Q, \tau) - H_{1,i}(\tilde{Q}, \tilde{\tau})\|_\infty \\ & \leq \beta_i \|Q - \tilde{Q}\|_\infty + \frac{L_i}{1 - \frac{\beta_i K_i}{2}} \left(\sum_{j=1}^N \|\tau_j - \tilde{\tau}_j\|_1 \right), \end{aligned}$$

and

$$\begin{aligned} & \|H_{2,i}(Q, \tau) - H_{2,i}(\tilde{Q}, \tilde{\tau})\|_1 \\ & \leq \frac{K_i}{\rho_i} \|Q - \tilde{Q}\|_\infty + \bar{K}_i \|\tau_i - \tilde{\tau}_i\|_1 + K_i \sum_{j \neq i} \|\tau_j - \tilde{\tau}_j\|_1. \end{aligned}$$

Proof: This result follows from [12, Lemma 2]. ■

III. A QUASI-STATIC CONTRACTION CONDITION FOR STATIONARY MFG

Let $(X, \mathcal{P}(A), (C_i + \rho_i \Omega_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$ be a regularized multi-population MFG, and suppose that the objective functional of each population $i \in [N]$ is discounted with factor $\beta_i < 1$. A quasi-static iteration scheme for discrete-time MFGs consists of iterating the state measures via the operators $H_{2,i}$ for each population $i \in [N]$, and then, under the updated state measures, computing the fixed point of $H_{1,i}$, which amounts to solving an MDP as an intermediate step. Obtaining a contraction condition in this setting is typically easier than in the case where Q -functions and state measures are updated consecutively at each iteration. We present a result in the stationary case to facilitate comparison with the asymptotic contraction condition we later derive for finite-horizon MFEs. The first step is to express the iterations as a vector inequality. For a given $\tau \in \prod_{i \in [N]} \mathcal{P}(X)$, we denote by $Q^{\tau,i}$ the fixed point of $H_{1,i}(\cdot, \tau)$.

Lemma 3: For all $\tau, \tilde{\tau} \in \prod_{i \in [N]} \mathcal{P}(X)$, we have

$$\begin{bmatrix} \|H_{2,1}(Q^{\tau,1}, \tau) - H_{2,1}(Q^{\tilde{\tau},1}, \tilde{\tau})\|_1 \\ \|H_{2,2}(Q^{\tau,2}, \tau) - H_{2,2}(Q^{\tilde{\tau},2}, \tilde{\tau})\|_1 \\ \vdots \\ \|H_{2,N}(Q^{\tau,N}, \tau) - H_{2,N}(Q^{\tilde{\tau},N}, \tilde{\tau})\|_1 \end{bmatrix} \leq \mathcal{M} \begin{bmatrix} \|\tau_1 - \tilde{\tau}_1\|_1 \\ \|\tau_2 - \tilde{\tau}_2\|_1 \\ \vdots \\ \|\tau_N - \tilde{\tau}_N\|_1 \end{bmatrix},$$

where $\mathcal{M} = [M_{i,j}]_{i,j}$ is such that $M_{i,j} = \bar{K}_i \mathbf{1}_{\{i=j\}} + K_i \mathbf{1}_{\{i \neq j\}} + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i)}$.

Proof: The result follows directly from Lemma 2; see also the proof of Lemma 4 in [12]. ■

A general eventual contraction result using the vector iterations above is the following:

Theorem 1: Let $\rho(\mathcal{M})$ be the spectral radius of $\mathcal{M}$. If $\rho(\mathcal{M}) < 1$, then the map $\tau \mapsto (H_{2,i}(Q^{\tau,i}, \tau))_{i \in [N]}$ is eventually contractive over $\prod_{i \in [N]} \mathcal{P}(X)$ under any norm.

Proof: This follows directly from the Gelfand's formula. ■

A closed-form expression for $\rho(\mathcal{M})$ is not always available, since it corresponds to an eigenvalue of $\mathcal{M}$. The next theorem provides an easier-to-check characterization of the condition $\rho(\mathcal{M}) < 1$ via an algebraic manipulation.

Theorem 2: We have $\rho(\mathcal{M}) < 1$ if and only if

$$\sum_{i \in [N]} \left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i)} \right) \frac{1}{(1 - (\bar{K}_i - K_i))} < 1$$

Proof: As a consequence of the Perron-Frobenius theorem, there exists a positive eigenvector $h = (h_i)_{i=1}^N$ for $\mathcal{M}$ that corresponds to $\rho(\mathcal{M})$. Then, using the equality of $\mathcal{M}h = \rho(\mathcal{M})h$ on the j 'th row, one can see that we have

$$(\bar{K}_j - K_j)h_j + \underbrace{\sum_{i \in [N]} \left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i)} \right) h_i}_{=k} = \rho(\mathcal{M})h_j,$$

which then implies that

$$k = \sum_{i \in [N]} \frac{\left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i)} \right)}{\rho(\mathcal{M}) - (\bar{K}_j - K_j)} h_i.$$

Consequently, we must have $\sum_{i \in [N]} \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i)})}{\rho(\mathcal{M}) - (\bar{K}_j - K_j)} = 1$. Note that the function

$$x \mapsto \sum_{i \in [N]} \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i)})}{x - (\bar{K}_j - K_j)} \quad (3)$$

is monotone decreasing on $(\max_j(\bar{K}_j - K_j), \infty)$; thus, there exists a unique $x > \max_j(\bar{K}_j - K_j)$ that satisfies $\sum_{i \in [N]} \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i)})}{\rho(\mathcal{M}) - (\bar{K}_j - K_j)} = 1$. It is straightforward to check that the function defined by (3) being less than 1 at $x = 1$ is equivalent to $\rho(\mathcal{M}) < 1$, as desired. ■

IV. VARIATIONAL CONTRACTION CONDITION FOR FINITE-HORIZON MFG

Similar to the stationary case, we will express our iterations purely in terms of iterations of state-measures.

Definition 2: Let $\tau = (\tau_t)_{t=0}^{T-1} \in \prod_{t=0}^{T-1} \left(\prod_{i=1}^N \mathcal{P}(X) \right)$. Then, by $(Q_t^{\tau,i})_{t=0}^{T-1}$, for all $i \in \{1, 2, \dots, N\}$, starting from the time step $t = T - 1$, we denote the Q -functions as

$$Q_{T-1}^{\tau,i}(x, u) = C(x, u, \tau_{T-1}) + \rho\Omega(u),$$

and

$$Q_t^{\tau,i}(x, u) = C(x, u, \tau_t) + \rho\Omega(u) + \beta \sum_{y \in X} Q_{\min, t+1}^{\tau,i}(y) P(y|x, u, \tau_t)$$

are defined recursively until $t = 0$. By $Q^{\tau,i} = (Q_t^{\tau,i})_{t=0}^{T-1}$ we will denote the flow generated by these Q -functions.

We are mainly interested in a horizon-length-independent eventual contraction condition in the finite-horizon case, in order to understand the large-time behavior of finite-horizon systems. Our main contraction condition is as follows:

Theorem 3: Let $\bar{K}_\infty = \min_j(\bar{K}_j - K_j)$ and $\beta_{\max} = \max_j \beta_j$. If

$$\inf_{\bar{K}_\infty < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i r)}}{1 - \frac{\bar{K}_i - K_i}{r}} < 1, \quad (4)$$

then, for every $T \in \mathbb{N}$, the map

$$\tau \mapsto \left((\tau_{0,i})_{i \in [N]}, H_{2,i}(Q_t^{\tau,i}, \tau_t) \right)_{i=1, t=1}^{N, T-1}$$

is eventually contractive on

$$\prod_{t=1}^{T-1} \left(\prod_{i=1}^N \mathcal{P}(X) \right) \times \prod_{i=1}^N \{\tau_{0,i}\}$$

with respect to any norm, for every initial state measure $(\tau_{0,i})_{i=1}^N \in \prod_{i=1}^N \mathcal{P}(X)$.

Proof: In the remainder of this section, we outline the main ideas of the proof. For the full proof, we refer to [12, Theorem 3]. ■

Fixed points of the mapping

$$\tau \mapsto \left((\tau_{0,i})_{i \in [N]}, (H_{2,i}(Q_t^{\tau,i}, \tau_t))_{i=1, \dots, N, t=1, \dots, T-1} \right)$$

indeed induce a finite-horizon MFE. A special case of Theorem 3 is the single-population setting, which recovers [7, Theorem 2] via a variational contraction condition.

Corollary 1: Suppose $N = 1$, i.e. we have a single population MFG. Then, we have eventual contraction for mean-field equilibrium operator if

$$\inf_{\bar{K}_\infty < r < \beta_{\max}^{-1}} \frac{\bar{K}_1}{r} + \frac{\bar{L}_1 K_1}{1 - \beta_1 r} < 1.$$

Proof: This result directly follows from Theorem 3. ■

As in the stationary case, the finite-horizon contraction result above is obtained by identifying a vector inequality for the iterations.

Lemma 4: For all $i \in [N]$, let $\bar{\mathcal{M}}_i, \mathcal{M}_i \in \mathbb{R}^{(T-1) \times (T-1)}$ be such that $\bar{\mathcal{M}}_i = T_i(\bar{K}_i)$ and $\mathcal{M}_i = T_i(K_i)$, where

$$T_i(x) = \begin{bmatrix} \frac{\bar{L}_i K_i}{\rho_i} \beta_i & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^2 & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^3 & \dots & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^{T-1} \\ x + \frac{\bar{L}_i K_i}{\rho_i} & \frac{\bar{L}_i K_i}{\rho_i} \beta_i & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^2 & \dots & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^{T-2} \\ 0 & x + \frac{\bar{L}_i K_i}{\rho_i} & \frac{\bar{L}_i K_i}{\rho_i} \beta_i & \dots & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^{T-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \frac{\bar{L}_i K_i}{\rho_i} \beta_i \end{bmatrix}.$$

By $S_{a,b}(\mathcal{M}_i, \mathcal{M}_j)$, denote an $a(T-1) \times b(T-1)$ dimensional block matrix with $\mathcal{M}_i$ on the diagonals and $\mathcal{M}_j$ on the off-diagonals. By $E_{i,j}$ we denote an $N \times N$ -dimensional matrix whose entries all zero except the row i and column j . We define $\mathcal{S}_T = \sum_{i=1}^N E_{i,i} \otimes S_{N_i, N_i}(\bar{\mathcal{M}}_i, \mathcal{M}_i) + \sum_{i,j:i \neq j} E_{i,j} \otimes S_{N_i, N_j}(\mathcal{M}_i, \mathcal{M}_j)$. For any $\tau, \tilde{\tau} \in \prod_{t=0}^{T-1} \left(\prod_{i=1}^N \mathcal{P}(X) \right)$ define $\delta_{i,t} := \|H_{2,i}(Q_t^{\tau,i}, \tau_t) - H_{2,i}(Q_t^{\tilde{\tau},i}, \tilde{\tau}_t)\|_1$, $\Delta_{i,t} := \|\tau_i(t) - \tilde{\tau}_i(t)\|_1$, $i \in [N]$, $t = 1, \dots, T-1$, $\delta := \sum_{i=1}^N \sum_{t=1}^{T-1} \delta_{i,t} (e_i \otimes e_t) \in \mathbb{R}^{N(T-1)}$, and $\Delta := \sum_{i=1}^N \sum_{t=1}^{T-1} \Delta_{i,t} (e_i \otimes e_t) \in \mathbb{R}^{N(T-1)}$. Then, we have $\delta \leq \mathcal{S}_T \Delta$.

Proof: This result is a straightforward application of Lemma 2. Also see [12, Lemma 5]. ■

The following theorem characterizes the eventual contraction of the mapping $\tau \mapsto ((\tau_{0,i})_{i \in [N]}, H_{2,i}(Q_t^{\tau,i}, \tau_t))_{i=1, \dots, N, t=1, \dots, T-1}$ by providing an easy-to-check condition ensuring that $\rho(\mathcal{S}_T) < 1$ for all T , thereby recovering Theorem 3.

Theorem 4: We have $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) < 1$ if and only if

$$\inf_{\bar{K}_\infty < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i r)}}{1 - \frac{\bar{K}_i - K_i}{r}} < 1. \quad (5)$$

Proof: Our uniqueness result relies only on the “if” part, so we provide a proof of this implication. The proof of the “only if” part is rather involved, and we therefore refer the reader to [12, Theorem 4].

For the “if” part, we apply the Collatz-Wielandt formula to $\mathcal{S}_T$ using geometric test vectors. Fix $0 < r < \beta_{\max}^{-1}$ and let $c = (c_1, \dots, c_N) \in \mathbb{R}_{>0}^N$. Define $x = (x_i)_{i=1}^{T-1} \in \mathbb{R}_{>0}^{T-1}$ by $x_i = r^{i-1}$. Further, let $C = \sum_{i \in [N]} c_i$ and $g = \sum_{i \in [N]} c_i x = Cx$. For each $i \in [T-1]$, and $i \geq 2$, for any $m \in [N-1]$ we have

$$(\bar{K}_m - K_m) c_m x_{i-1} + C \left(K_m + \frac{\bar{L}_m K_m}{\rho_m} \right) x_{i-1}$$

$$+ C \sum_{j=i}^{T-1} \beta_m^{j-i+1} x_j = (\mathcal{S}_T g)_{m(T-1)+i};$$

which implies that

$$\begin{aligned} \frac{\bar{K}_m - K_m}{r} + \frac{C \left(K_m + \frac{\bar{L}_m K_m}{\rho_m} \right)}{c_m r} + \frac{C \bar{L}_m K_m \beta_m}{\rho_m c_m (1 - \beta_m r)} \\ \geq \frac{(\mathcal{S}_T g)_{m(T-1)+i}}{c_m x_i}. \end{aligned}$$

Let $B(r)$ be the matrix $B(r) := D(r) + \mathbf{1} v(r)^\top$, where $\mathbf{1}$ is the vector of ones and $v_j(r) = \frac{K_j}{r} + \frac{\bar{L}_j K_j}{\rho_j(1-\beta_j r)}$, $D(r) := \text{diag} \left(\frac{\bar{K}_1 - K_1}{r}, \frac{\bar{K}_2 - K_2}{r}, \dots, \frac{\bar{K}_N - K_N}{r} \right)$, and $v(r) := (v_1(r), \dots, v_N(r))^\top$. With this notation, from the argument above it follows that $\frac{(\mathcal{S}_T g)_{m(T-1)+i}}{c_m x_i} \leq \frac{(B(r) \mathbf{c}^\top)_m}{c_m}$. Collatz-Wielandt formula then implies $\rho(\mathcal{S}_T) \leq \rho(B(r))$ for all $\beta_{\max}^{-1} > r > 0$. Note that $(\rho(\mathcal{S}_T))_T$ is a monotone subsequence; thus, $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) \leq \inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r))$. After repeating the proof of Theorem 2 for $\rho(B(r))$, one can easily obtain that $\inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r)) < 1$ if and only if $\inf_{0 < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\left(\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i r)} \right)}{1 - \frac{\bar{K}_i - K_i}{r}} < 1$. Infimum over $\bar{K}_\infty < r < \beta_{\max}^{-1}$ also comes from the “only if” direction, so we omit the details for this part. ■

For the “only if” part, the proof used in the stationary case does not scale to the finite-horizon setting. The main idea behind the “only if” part of the proof of Theorem 4 is to characterize the positive fixed points of the limiting operator “ $\lim_{T \rightarrow \infty} \mathcal{S}_T$ ”, which is an infinite-dimensional operator. The characterization above is then obtained using analytic-combinatoric methods.

The proof of the “only if” direction of Theorem 4 proceeds as follows. In order to apply analytic-combinatorial methods, we first show that the limiting operator “ $\lim_{T \rightarrow \infty} \mathcal{S}_T$ ” admits a positive fixed point. To do so, we prove that the normalized eigenvectors of the matrices $\mathcal{S}_T$ satisfy sufficient regularity properties to pass to the limit; see [12, Proposition 2].

Let $(u_i(k))_{i \in [N], k \in \mathbb{N}}$ be a positive fixed point of $\lim_{T \rightarrow \infty} \mathcal{S}_T$. We then show that, for each $i \in [N]$,

$$\lim_{k \rightarrow \infty} \frac{u_i(k+1)}{u_i(k)} = c$$

for some constant c , and that, for all $i \neq j$,

$$\lim_{k \rightarrow \infty} \frac{u_i(k)}{u_j(k)} = \frac{c_i}{c_j}$$

for some constants $c_i, c_j > 0$. These asymptotic ratio estimates are then used to prove that the spectral radius of the limiting operator $\lim_{T \rightarrow \infty} \mathcal{S}_T$ is equal to $\rho(B(1/c))$; see [12, Lemma 19].

If $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) < 1$, then part of the proof of Theorem 4 implies that $\bar{K}_\infty < r < \beta_{\max}^{-1}$. Thus, when (5) holds with $r = 1$, one recovers the contraction condition for stationary MFGs; see Theorem 2. In this sense, the contraction condition for finite-horizon MFEs is weaker than that for stationary MFGs.

Let

$$V(r) := \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i(1-\beta_i r)}}{1 - \frac{\bar{K}_i - K_i}{r}}.$$

We call V a *variational contraction function* for the finite-horizon MFE. The quantity $\inf_r V(r)$ captures the *worst-case* behavior of the finite-horizon MFG that can be deduced from a geometric majorization of $\mathcal{S}_T$ through the matrices $B(r)$. It can be shown that $\inf_{\bar{K}_\infty < r < \beta_{\max}^{-1}} \rho(B(r)) < 1$ if and only if $\inf_{\bar{K}_\infty < r < \beta_{\max}^{-1}} V(r) < 1$; see [12, Lemma 21]. As shown in the proof of Theorem 4, the quantity $\rho(B(r))$ serves as a majorant for $\rho(\mathcal{S}_T)$, where $B(r)$ is used as an intermediate majorant for the matrices $\mathcal{S}_T$ via the Collatz-Wielandt formula. Since $\inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r)) < 1$ if and only if $V(r) < 1$ for some $r < 1$, the behavior of V determines when there would exist a contractive majorant for $\mathcal{S}_T$. Motivated by this observation, we distinguish three regimes: $r < 1$, $r = 1$, and $r > 1$.

- 1) When $V(1) < 1$, there exists a geometric majorization for $\mathcal{S}_T$ such that the ratios of consecutive terms in the positive Perron right eigenvectors converge to the constant 1. In this case, we say that $(\mathcal{S}_T)_T$ is *asymptotically stationary*, since we recover Theorem 2.
- 2) When $V(r) < 1$ for some $r > 1$, there exists a geometric majorization for $\mathcal{S}_T$ such that the ratios of consecutive terms in the positive Perron right eigenvectors converge to the constant r . In this case, we say that $(\mathcal{S}_T)_T$ is *stable*. This stability stems from the fact that the resulting majorization has eigenvectors that decay geometrically with ratio $1/r$. When $(\mathcal{S}_T)_T$ is asymptotically stationary, it is also stable.
- 3) When $V(r) < 1$ only for some $r < 1$, there exists a geometric majorization for $\mathcal{S}_T$ such that the ratios of consecutive terms in the positive Perron right eigenvectors converge to the constant r . In this case, we say that $(\mathcal{S}_T)_T$ is *unstable*.

In the next section, we show that one can obtain finite-time error bounds between finite-horizon and infinite-horizon MFEs when $\mathcal{S}_T$ is stable.

V. FINITE-HORIZON TO INFINITE-HORIZON: CONVERGENCE RATES AND UNIQUENESS

Let $(X, \mathcal{P}(A), (C_i + \rho_i \Omega_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$ be a regularized multi-population MFG, in which the objective of each population i is discounted by the factor $\beta_i < 1$. In this section, we study the convergence of finite-horizon MFEs to the infinite-horizon MFE in the multi-population setting.

For any $(\pi^\infty, \tau^\infty) \in \text{MFE}_{\tau_0}$, we denote by $(\pi^\infty|_T, \tau^\infty|_T)$ the truncation of this infinite-horizon non-stationary MFE to the first T time steps for each population. The main result of this section is the following finite-time error bound between a finite-horizon MFE and an infinite-horizon non-stationary MFE, from which uniqueness of the non-stationary MFE follows directly:

Theorem 5: Let $(\pi^T, \tau^T) \in \text{MFE}_{T, \tau_0}$. Suppose that there exists $t_0 > 1$ such that $V(t_0) < 1$. Then, for all $t \ll T$, we have

$$\sum_{i \in [N]} \|\tau_{t,i}^\infty|_T - \tau_{t,i}^T\|_1 \leq O\left(\frac{1}{t_0^{T-t}}\right).$$

In particular, $|\text{MFE}_{\tau_0}| = 1$, i.e., there exists a unique non-stationary MFE in the N -population case for any given family of initial state measures τ_0 .

Proof: Define $\bar{t}_* = (t_0^{T-2}, t_0^{T-3}, \dots, t_0, 1)$ and let $c = (c_i)_i$ be a positive left Perron eigenvector of the matrix $B(t_0)$ that we used in the proof of Theorem 4. We further define $\hat{t}_* = (c_1 \bar{t}_*, c_2 \bar{t}_*, \dots, c_N \bar{t}_*)$. Note that

$$\frac{(\mathcal{S}_T^* \hat{t}_*)_{m(T-1)+i}}{c_m \bar{t}_{*,i}} \leq \rho(B(t_*)),$$

see [12, Lemma 22] for details. This inequality implies that for any positive vector $y \in \mathbb{R}^{N(T-1)}$ we have

$$\langle \hat{t}_*, \mathcal{S}_T y \rangle \leq \rho(B(t_*)) \langle \hat{t}_*, y \rangle,$$

see [12, Lemma 23].

As a consequence of Lemma 2, we obtain

$$\begin{aligned} & \langle \hat{t}_*, (\|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1)_{t=1,j=1}^{T-1,N} \rangle \\ & \leq \langle \hat{t}_*, \mathcal{S}_T (\|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1)_{t=1,j=1}^{T-1,N} \rangle \\ & \quad + \langle \hat{t}_*, \bar{\mathcal{S}} \left(\underbrace{\|Q_T^{\tau^\infty, i}\|_\infty, \|Q_T^{\tau^\infty, i}\|_\infty, \dots, \|Q_T^{\tau^\infty, i}\|_\infty}_{T-1 \text{ times}} \right)_{i \in [N]} \rangle \\ & \leq \rho(B(t_*)) \langle \hat{t}_*, (\|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1)_{t=1,j=1}^{T-1,N} \rangle \\ & \quad + \langle \hat{t}_*, \bar{\mathcal{S}} (\|Q_T^{\tau^\infty, i}\|_\infty, \|Q_T^{\tau^\infty, i}\|_\infty, \dots, \|Q_T^{\tau^\infty, i}\|_\infty)_{i \in [N]} \rangle, \end{aligned}$$

where $\bar{\mathcal{S}} = \begin{bmatrix} \tau_1 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \tau_2 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \tau_N \end{bmatrix}$, $\tau_i = \begin{bmatrix} \beta_i^T & 0 & \dots & 0 \\ 0 & \beta_i^{T-1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \beta_i \end{bmatrix}$, and $\mathbf{0}$ is a $(T-1) \times (T-1)$

dimensional matrix made of 0's. Here, the terms $\|Q_T^{\tau^\infty, i}\|_\infty$ arise from the fact that τ^∞ induces an infinite-horizon MFE. This follows directly from Lemma 2 by recursive iteration, as in the proof of Lemma 4.

Since $V(t_0) < 1$, we can rearrange the inequality above to obtain that

$$\begin{aligned} & \min_i c_i t_*^{T-t} \sum_{j \in [N]} \|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1 \\ & \leq \langle \hat{t}_*, (\|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1)_{t=1,j=1}^{T-1,N} \rangle \\ & \quad \underbrace{\langle \hat{t}_*, \bar{\mathcal{S}} (\|Q_T^{\tau^\infty, i}\|_\infty, \|Q_T^{\tau^\infty, i}\|_\infty, \dots, \|Q_T^{\tau^\infty, i}\|_\infty)_{i \in [N]} \rangle}_{=I} \\ & \leq \frac{I}{1 - \rho(B(t_*))}. \end{aligned}$$

It remains to control the term I . Let $c_\infty = \max_i c_i$, and $\mathcal{T}_{\arg \max_i \beta_i} = \mathcal{T}_\infty$. First, since $t_0 \beta_{\max} < 1$, we have

$$\begin{aligned} I & \leq N c_\infty \langle \bar{t}_*, \mathcal{T}_\infty \left(\underbrace{\max_i \|Q_T^{\tau^\infty, i}\|_\infty, \dots, \max_i \|Q_T^{\tau^\infty, i}\|_\infty}_{T-1 \text{ times}} \right) \rangle \\ & \leq N c_\infty \max_i \|Q_T^{\tau^\infty, i}\|_\infty \beta_{\max}^2 t_o \frac{\beta_{\max}^{T-1} t_o^{T-1} - 1}{\beta_{\max} t_o - 1}. \end{aligned}$$

Consequently, for the differences at stage t across all population groups, and using the fact that $c_i > 0$ for all $i \in [N]$, we obtain

$$\begin{aligned} & \sum_{j \in [N]} \|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1 \\ & \leq \frac{N \frac{c_\infty}{\min_i c_i} \max_i \|Q_T^{\tau^\infty, i}\|_\infty}{1 - \rho(B(t_*))} \beta_{\max}^2 t_o^{t+1} \frac{\beta_{\max}^{T-1} t_o - \frac{1}{t_o^{T-t}}}{\beta_{\max} t_o - 1}. \end{aligned}$$

Since $\beta_{\max} < 1$ and $t_o > 1$, it follows that

$$\sum_{j \in [N]} \|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1 \rightarrow 0 \quad \text{as } T \rightarrow \infty.$$

Moreover, because $t_o \beta_{\max} < 1$, the dominant term in the above expression is $\frac{1}{t_o^{T-t}}$, which yields the desired error bound.

It remains to show the existence and uniqueness of an infinite-horizon non-stationary MFE under a given initial state-measure. In the error bounds derived above, the state-measure flow τ^∞ was chosen arbitrarily among the infinite-horizon MFEs. Let $\tilde{\tau}^\infty$ be another state-measure flow corresponding to an infinite-horizon MFE with the same initial state measure as τ^∞ . Then the preceding argument shows that any finite-horizon MFE with this initial state measure also converges to $\tilde{\tau}^\infty$. By uniqueness of limits in $\prod_{t=0}^\infty \mathcal{P}(X)$, where each $\mathcal{P}(X)$ is endowed with the topology of weak convergence, it follows that $\tilde{\tau}^\infty = \tau^\infty$. Due to the presence of the regularizer, each state-measure flow τ^∞ has a unique corresponding policy flow; thus, there exists a unique infinite-horizon non-stationary MFE starting from any initial-state measure, as desired. ■

The proof of the result above relies on an extension of Lemma 4, taking into account that the truncated components $(\pi^\infty|_T, \tau^\infty|_T)$ induce Q -functions with a nontrivial terminal condition. The matrix $\mathcal{S}_T$ is then majorized using appropriate values of $V(r)$, under a suitable choice of geometric test vectors.

Remark 1: If $V(1) < 1$, then the conditions of Theorem 5 are satisfied. If $V(1) > 1$ and $V(t_*) > 1$, then Theorem 5 is not applicable. Thus, the interesting case is when $V(1) > 1$ and $V(t_*) < 1$.

We also note that when $N = 1$, Theorem 5 remains effective whenever the stationary MFE is contractive, a case that cannot be recovered using only the worst-case considerations in [7, Theorem 4]. Moreover, when the stationary MFG is contractive, we can establish a finite-time error bound between the infinite-horizon non-stationary MFE and the stationary MFE.

Theorem 6: Suppose that there exists $t_* > 1$ such that $V(t_*) < 1$. Let (π, τ) be a stationary MFE and $(\pi^\infty, \tau^\infty) \in \text{MFE}_{\tau_0}$. Then, it holds that

$$\begin{aligned} & \sup_{i \in [N], t \geq k} \|\tau_{t,i}^\infty - \tau_i\|_1 \\ & \leq O \left(\max_j \left(\frac{\bar{L}_j K_j}{\rho_j(1 - \beta_j)} + \bar{K}_j \right)^k \right). \end{aligned}$$

Proof: See [12, Theorem 7]. ■

Theorem 6, combined with Theorem 5, yields a finite-time error bound between the stationary MFE and the finite-horizon MFE whenever the stationary MFG is contractive, via a straightforward application of the triangle inequality.

Corollary 2: Let (π, τ) be a stationary MFE and $(\pi, \tau) \in \text{MFE}_{T, \tau_0}$. Suppose $\bar{K}_\infty < t_* < \beta_{\max}^{-1}$ is such that $V(t_*) < 1$. Then, for all $k \geq 0$ we have that

$$\|\tau_{k,i} - \tau_i\|_1 \leq O \left(\left(\frac{\max_j (K_j + \frac{\bar{L}_j K_j}{\rho_j})}{1 - \rho(B(t_*))} \right)^k + \frac{1}{t_*^{T-k}} \right)$$

Proof: This follows from Theorem 6 and Theorem 5 using a straightforward triangle inequality. ■

VI. CONCLUSION

In this paper, we have presented contraction conditions for multi-population discrete-time regularized MFGs. Using the variational nature of these contraction conditions, we have obtained error bounds between finite-horizon and infinite-horizon MFE.

The variational contraction conditions we have presented captures the asymptotics of the spectral radius that correspond to the contraction matrix in our setting, and provides a more general working conditions under which one can obtain error bounds between finite-horizon and infinite-horizon MFE compared to the one given in [7]. However, our method does not provide a tight lower bound for the spectral radius in the finite-horizon setting unlike [7, Theorem 2]. Our numerical results suggest that the spectral radius in the multi-population case should admit a lower bound similar to the one derived in [7]. Establishing a tight lower bound for $\rho(\mathcal{S}_T)$ is an important direction for future research, as it would provide further insight into the asymptotic behavior of contraction rates for finite-horizon MFE in the multi-population setting.

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